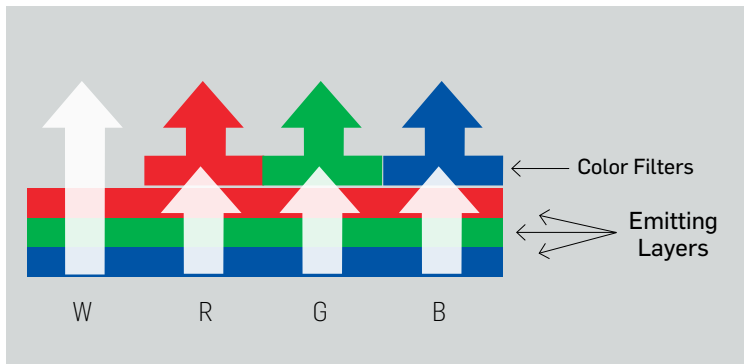


additional white sub-pixel for boosting the brightness. Thus, you have a WRGB arrangement. LG claims that this arrangement and mixture of colours makes manufacturing OLEDs easier. Along with the claimed long life of the panels, WRGB arrangement also helps in cutting costs, making its production much easier and less expensive.

Advantages/Pros:

An OLED display checks the right boxes to offer various advantages over the older, traditional panels. Some of them are listed below.

- ◆ Starting off with the physical advantages, OLEDs are about 0.2 mm to 0.3 mm thick whereas LCDs are at least 10 times thicker.
- ◆ The OLED have plastic substrates on both sides which reduces weight and makes it flexible than other display panels, resulting in manufacturing thin and large sheets of OLED easily.
- ◆ OLEDs are much brighter than regular LEDs as the organic layer is much thinner as compared to the other layers in a native LED display.
- ◆ It also does not need any backlighting as OLEDs generate lights themselves which reduces power consumption and provides wider viewing angles, even when the viewing angle approaches 90-degrees from the normal surface.
- ◆ It is possible for an OLED to be partially or completely off, thus reproducing deep blacks and dark greys easily, which is impossible to achieve in LCDs due to the backlighting it requires.
- ◆ OLED displays can also achieve higher contrast ratio in dynamic as well



LG's WRGB arrangement in its OLED panels